

HPATG CERTIFICATION

01

THE ANCHOR MODULE

FOUNDATIONAL FRAMEWORK

A SYSTEMS-BIOLOGY APPROACH TO HORMONES

Why the HPATG Axis is the key to functional endocrinology —
and the lens through which every module in this course is
understood.

HYPOTHALAMUS

PITUITARY

ADRENALS

THYROID

GONADS

+ INSULIN

MODULE 1

WHAT YOU WILL LEARN

By the end of this module, you will be equipped to apply the HPATG framework in clinical practice.

LEARNING OBJECTIVES

- 01** Explain the HPATG model and why treating hormones in isolation produces incomplete clinical results
- 02** Identify the three most common patterns of HPATG breakdown and the clinical picture each produces
- 03** Apply the three laws of hormones (Hierarchy, Goldilocks, Pleiotropy) to clinical decision-making
- 04** Use the 4-step clinical strategy (Brain First → Filter → Hierarchy → Adaptation) to sequence your approach
- 05** Recognize why women's hormone physiology requires a cycle-informed, performance-aware clinical lens
- 06** Understand how insulin integrates into the HPATG model as a cross-axis disruptor affecting all three downstream axes

MODULE 1

A SYSTEMS-BIOLOGY APPROACH TO HORMONES

Moving Beyond the "Single Hormone" Theory

THE PARADIGM SHIFT

THE OLD WAY

- ✗ **Symptom-Based:** Treating the complaint, not the cause.
- ✗ **Isolated Organs:** "Thyroid problem" vs. "Adrenal problem."
- ✗ **Reference Ranges:** "Normal for age" (statistical average).
- ✗ **Pharmacology First:** Drugs to suppress symptoms.

THE SYSTEMS WAY

- ✓ **Root-Cause Based:** Asking "Why is this happening?"
- ✓ **Interconnected Web:** The HPATG Axis as one system.
- ✓ **Optimal Ranges:** "Optimal for function" (physiologic peak).
- ✓ **Physiology First:** Restoring function before suppressing it.

"We don't treat the lab number. We treat the person attached to it."

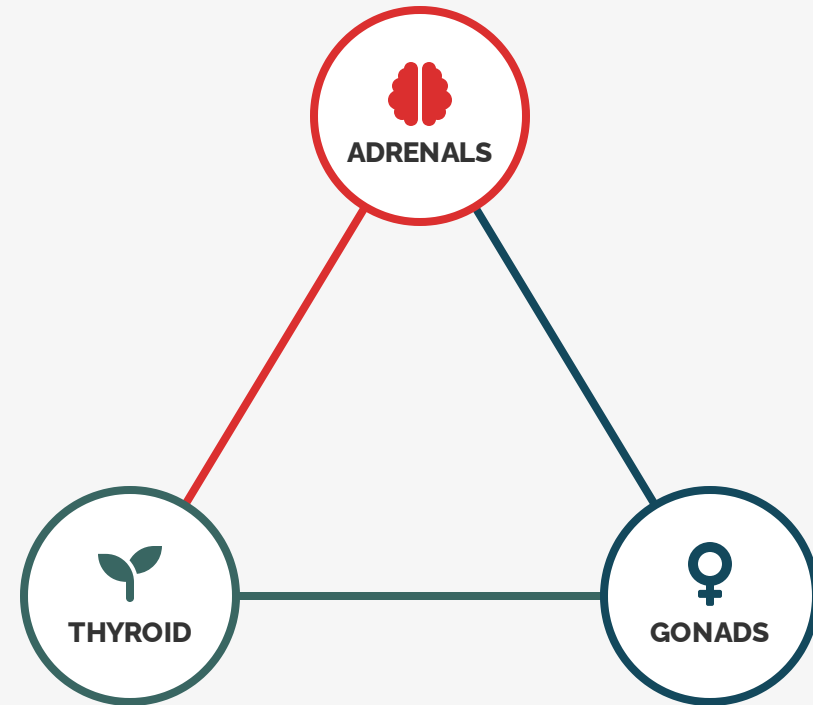
THE WEB OF HORMONES

THE MISTAKE

Treating the Thyroid, Adrenals, or Ovaries in isolation. You cannot pull one string without moving the others.

THE REALITY

High Cortisol (Adrenals) suppresses TSH (Thyroid) and blocks Ovulation (Gonads). We must assess the entire web.



WHY THIS LOOKS DIFFERENT IN WOMEN

- The menstrual cycle adds a 4th dimension: hormone levels shift dramatically across phases, meaning HPA reactivity and thyroid output vary by cycle day

- Perimenopause is the most significant HPATG breakdown event of a woman's lifetime — a full-axis reconfiguration, not just sex hormone decline

- In active or underfueling women, the primary HPA stressor is energetic, not psychological — requiring a completely different clinical approach

THE HPATG AXIS

1

HYPOTHALAMUS

The Sensor (CRH, TRH, GnRH)



2

PITUITARY

The Messenger (ACTH, TSH, LH/FSH)



3

GLANDS (A-T-G)

The Workers (Cortisol, T4, Estrogen)

THE CORPORATE METAPHOR



The CEO (Hypothalamus)

Senses the market (stress, light, temp) and makes the executive decision.



The Manager (Pituitary)

Sends the memo to the specific department.



The Factory (Glands)

Produces the actual product based on orders.

In functional presentations, dysfunction most often originates upstream — at the hypothalamus or pituitary. However, always rule out primary gland pathology first (Hashimoto's, primary adrenal insufficiency, ovarian failure). The framework guides your thinking — it doesn't replace a thorough workup.

THE 3 LAWS OF HORMONES



1. HIERARCHY

Survival always trumps Reproduction.
The body will "steal" resources from sex hormones to make stress hormones.

Example: High Cortisol = Low Progesterone



2. GOLDBLOCKS

More is not better. Hormones must be "just right." Too high causes resistance; too low causes deficiency.

Example: Insulin Resistance (High Insulin = No Effect)



3. PLEIOTROPY

One hormone has many jobs.
Estrogen isn't just for reproduction; it protects the brain, bones, and heart.

Example: Menopause = Brain Fog + Bone Loss

"Understanding these laws allows you to predict symptoms before labs even come back."

THE CLINICAL STRATEGY

1



BRAIN FIRST

Address the top of the axis. Circadian rhythm, sleep quality, and stress perception are the master switches. You cannot fix what is downstream if the brain is still in "alarm mode."

2



FILTER

Remove the interference. Is it toxins? Chronic inflammation? Poor sleep? Before adding hormones, remove what is blocking them.

3



HIERARCHY

Fix the order. Adrenals first, then Thyroid, then Sex Hormones. You cannot build a roof (Gonads) on a shaky foundation (Adrenals).

4



ADAPTATION

Build resilience. Use adaptogens, lifestyle, and hormetic stress to make the system stronger — not just dependent on support.

"Don't just add. Subtract first. And always start at the top."

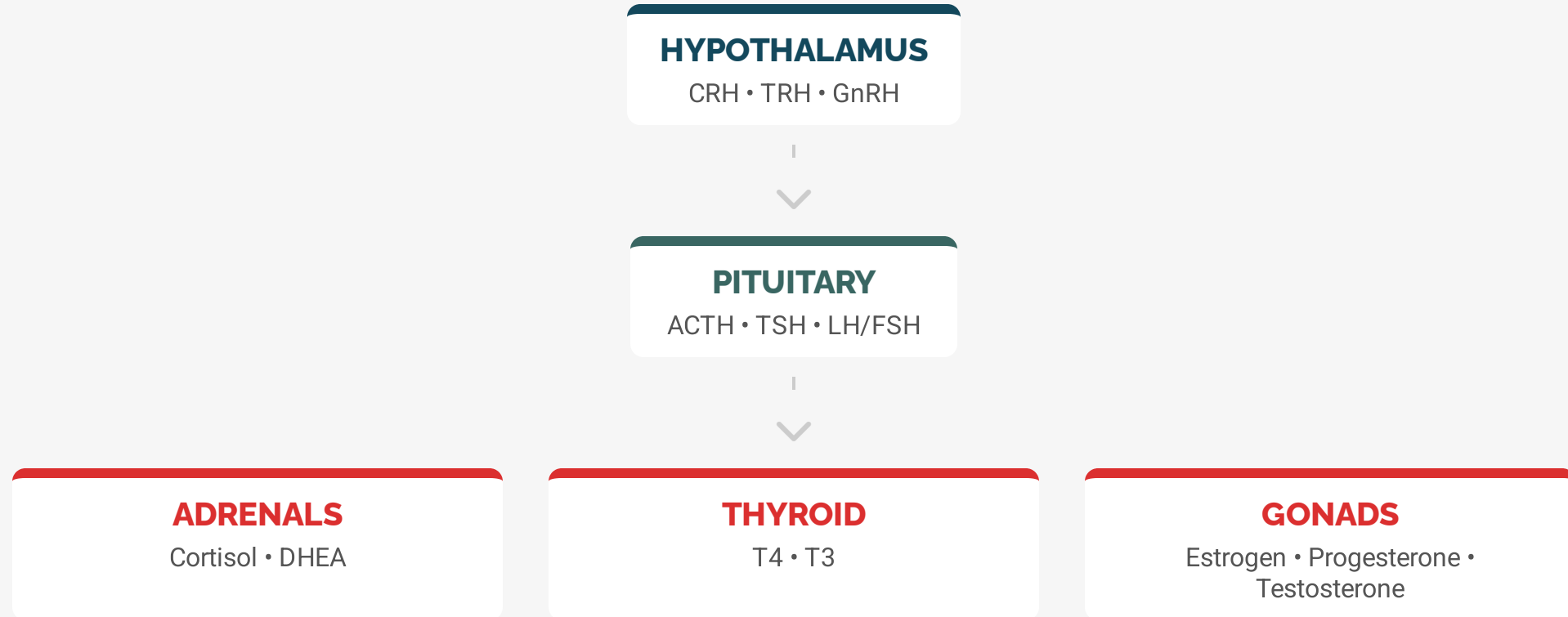
MODULE 1

MINI-MODULE 1B

THE HPATG AXIS

Mapping the Master Control System:
Hypothalamus, Pituitary, Adrenals, Thyroid, Gonads

THE HPATG MAP



+ **INSULIN: THE 4TH AXIS**
Insulin / Metabolic Core

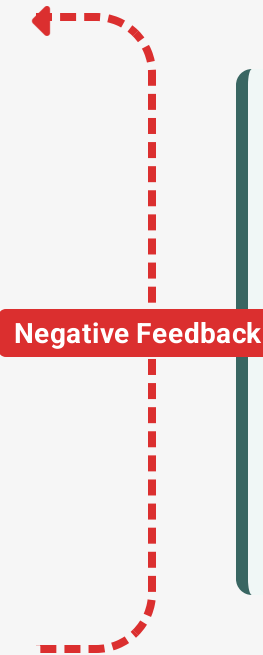
HPG: High insulin drives ovarian androgen production (PCOS mechanism) | **HPT:** Insulin resistance impairs T4 → T3 conversion | **HPA:** Blood sugar instability triggers cortisol output. You cannot optimize the HPATG axes without addressing the metabolic foundation.

FEEDBACK LOOPS

Brain (H-P)
Sends Stimulating Hormone



Gland (Target)
Produces Active Hormone



THE THERMOSTAT METAPHOR

Think of your home thermostat. When the house gets **hot enough** (high hormone levels), the AC **turns off** (brain stops signaling).

If the house is **too cold** (low hormone levels), the AC **kicks on** (brain sends more signal).

"High TSH means the brain is shouting because the thyroid isn't listening."

THE ONE EXCEPTION: THE LH SURGE

There is one critical exception to the negative feedback rule — and it happens only in women. As estrogen rises through the follicular phase, it eventually reaches a threshold that **flips the feedback signal from negative to positive**: instead of suppressing LH, high estrogen triggers a massive LH surge that causes ovulation. When this positive feedback loop fails — estrogen rises but the LH surge doesn't happen — the result is an anovulatory cycle: no ovulation, no corpus luteum, no progesterone.

CONVERSION CASCADES

T4

INACTIVE

LIVER / GUT



T3

ACTIVE

Testosterone

AROMATASE ENZYME



Estradiol

POTENT

PREGNENOLONE — THE MASTER PRECURSOR: 3 PATHWAYS

→ CORTISOL

Stress priority pathway — always first in line under chronic HPA activation

→ DHEA → Testosterone → Estradiol

Sex hormone pathway — sacrificed when cortisol demand is high

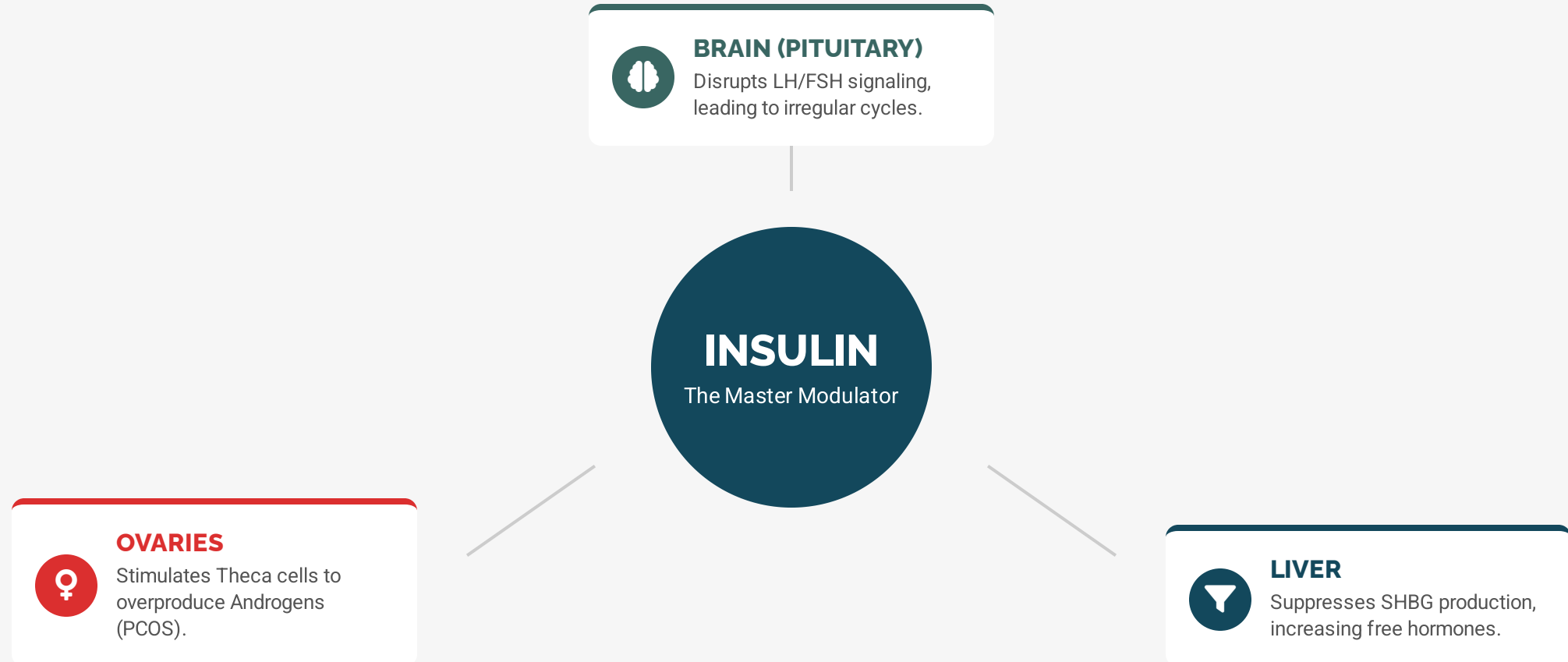
→ PROGESTERONE

Calming, anti-inflammatory pathway — also depleted under chronic stress

Under chronic stress, the body prioritizes cortisol production — leaving less pregnenolone available for DHEA and progesterone. This is why chronically stressed women experience low libido, irregular cycles, and poor sleep simultaneously.

"It's not just what you make. It's what you convert."

METABOLIC SIGNALS



"You cannot fix sex hormones without fixing insulin first."

KEY TAKEAWAYS



THE MAP IS REAL

The HPATG axis is not a theory. It is the physiological reality of how hormones are regulated.



LOOPS MATTER

Negative feedback loops keep the system in check. When they break, symptoms arise.



CONVERSION IS KEY

Making the hormone is only step one. Converting it to the active form is where the magic happens.



CLINICAL COMPASS

"When in doubt, zoom out. Look at the whole axis, not just the single hormone."

"Next: We will explore what happens when this system breaks down."

MODULE 1

MINI-MODULE 1C

WHEN THE HPATG AXIS BREAKS DOWN

Pattern Recognition: Identifying the 3 Most Common Breakdown
Patterns in Clinical Practice

THE BRAIN-LEVEL DRIVER

THE TOP-DOWN CASCADE



Limbic System

(Amygdala perceives threat)

Hypothalamus — CRH hypersecretion

Alarm signal — continuous, not pulsatile

Pituitary — ACTH overproduction → receptor downregulation

Corticotrophs desensitize over time

Adrenals — Initially ↑ cortisol → eventually blunted

Gland responds to a broken signal

Thyroid • Gonads • Metabolism

All downstream systems disrupted

1

CRH HYPERSECRETION

The brain perceives threat when it shouldn't. **Chronic stress, trauma, and sleep loss** keep the amygdala on low alarm, driving continuous CRH release instead of normal pulses.

ROOT CAUSE: PERCEIVED STRESS > ACTUAL STRESS

2

PITUITARY DESENSITIZATION

Sustained ACTH stimulation makes corticotrophs **downregulate CRH receptors**. The hypothalamus signals more; the pituitary responds less — producing blunted ACTH despite high CRH.

LAB CLUE: BLUNTED ACTH RESPONSE

3

CENTRAL FEEDBACK RESISTANCE

Chronic stress can make hippocampal glucocorticoid receptors less sensitive so cortisol no longer shuts the alarm off. The axis stays active despite circulating cortisol.

LAB CLUE: HIGH CORTISOL + CONTINUED HPA ACTIVATION



Clinical Pearl: The glands are often innocent bystanders. When the brain is in alarm mode, glandular support alone won't resolve the pattern — address the top of the axis first.

PATTERN 1: THE THYROID STEAL

THE MECHANISM

High Cortisol (Stress)



T4 (Inactive)



BLOCKS CONVERSION

Low T3 (Active)

CLINICAL PRESENTATION

- ⚠ **Fatigue:** "Tired but wired."
- ⚠ **Hair Loss:** Diffuse thinning.
- ⚠ **Cold Intolerance:** Cold hands/feet.
- ⚠ **Weight Gain:** Despite diet/exercise.

Lab Trap: TSH and T4 often look "normal." You must test Free T3 and Reverse T3 to see this pattern.

"The thyroid is the victim. The adrenal is the culprit."

PATTERN 2: THE INSULIN DRIVER

THE MECHANISM

High Insulin (Sugar/Carbs)



Stimulates Theca Cells

Ovaries



Overproduce

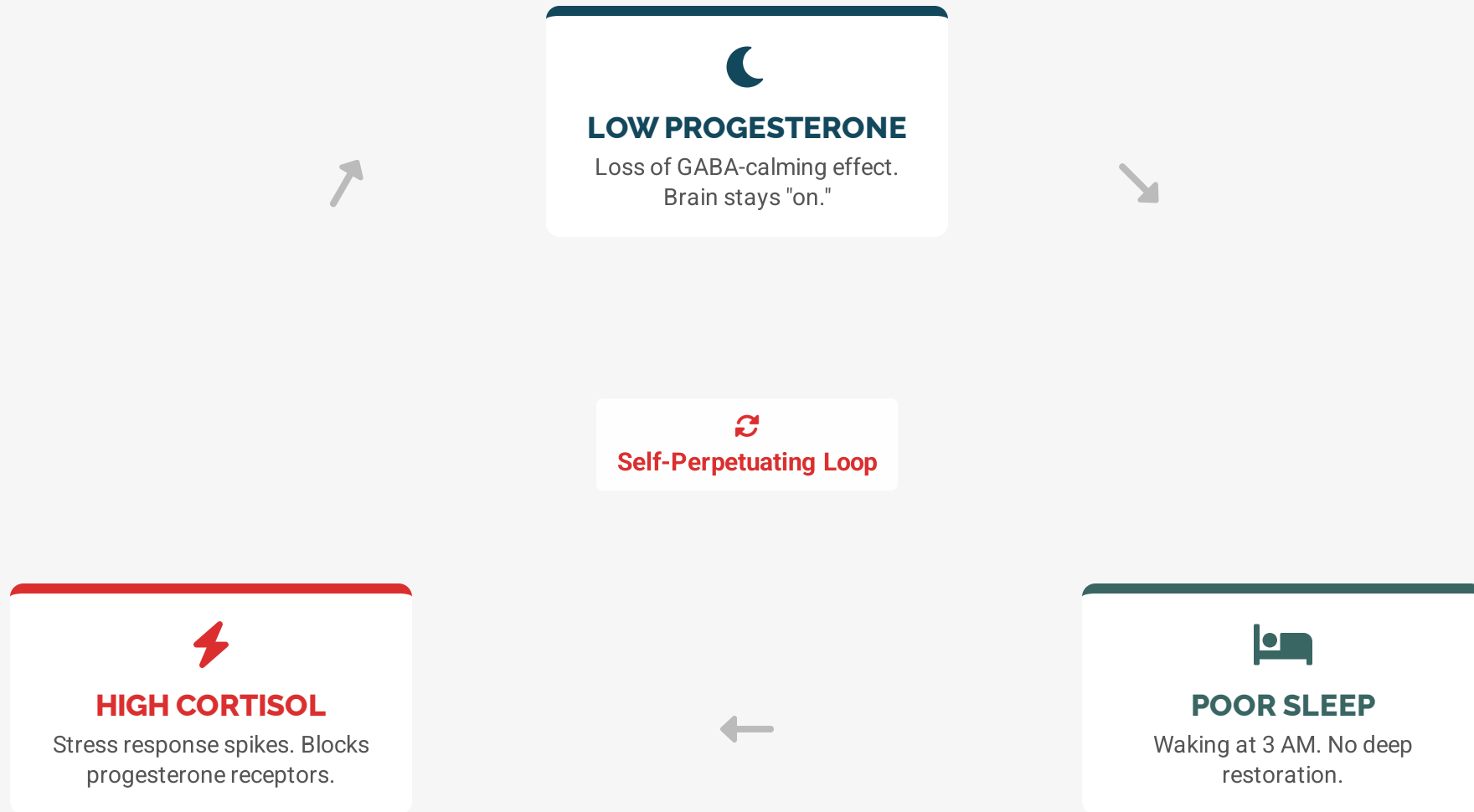
High Testosterone

CLINICAL PRESENTATION

- ! **Cystic Acne:** Jawline and chin.
- ! **Hirsutism:** Dark hair on face/chest.
- ! **Irregular Cycles:** Long cycles (>35 days).
- ! **Mood Swings:** "Hangry" episodes.

***Lab Trap:** Testosterone can be normal if SHBG is low.
Check Free Testosterone and DHEA-S.*

PATTERN 3: THE VICIOUS CYCLE



"You can't sleep because of hormones. Your hormones are wrecked because you can't sleep."

PATTERN RECOGNITION

PATTERN	MECHANISM	KEY SYMPTOMS
THYROID STEAL	High Cortisol impairs T4 to T3 conversion. <i>"Functional Hypothyroidism"</i>	FatigueHair LossCold
INSULIN DRIVER	High Insulin stimulates ovarian androgen production. <i>"Metabolic PCOS"</i>	AcneHirsutismIrregular Cycles
SLEEP STEAL	Low Progesterone reduces GABA. Poor sleep spikes Cortisol. <i>"The Vicious Cycle"</i>	InsomniaAnxiety3 AM Waking

"Recognize the pattern, and you find the root cause."

KEY TAKEAWAYS



CORTISOL STEALS

Stress hormones steal resources from thyroid and sex hormones. Always check adrenals first.



INSULIN DRIVES

High insulin drives high androgens. PCOS is often a sugar problem, not just an ovary problem.



CYCLES PERPETUATE

Symptoms (like insomnia) can become causes (high cortisol), creating self-sustaining loops.

PATTERN RECOGNITION

*"Don't just treat the symptom.
Break the cycle."*

"Next: How to translate this into testing and treatment."

CONVENTIONAL WORKUP

38-year-old female. Chief complaints: fatigue, hair loss, can't lose weight, irregular cycles, PMS.

TSH	2.4 mIU/L — "normal"
Estradiol Day 3	Mid-range — "normal"
Fasting Glucose	92 mg/dL — "normal"

"Labs are fine. Probably stress or perimenopause."

HPATG LENS REVEALED

4-pt Salivary Cortisol	Elevated AM, flat PM
HPA dysfunction, Stage 1	
Free T3	2.2 pg/mL — Low
Cortisol impairing T4 → T3 conversion	
Reverse T3	22 ng/dL — Elevated
T4 shunting to inactive form	
Mid-Luteal Progesterone	4.2 ng/mL — Insufficient
Anovulatory cycles from HPA suppression of HPG axis	
Fasting Insulin	14 µIU/mL — Elevated
Insulin resistance driving SHBG suppression & androgen excess	

ROOT CAUSE

HPA dysfunction — driven by chronic stress and blood sugar instability — was simultaneously suppressing thyroid conversion, blocking ovulation, and worsening insulin sensitivity. Treating the thyroid alone, or the hormones alone, would have failed. Stabilize blood sugar → support HPA → thyroid and HPG normalize downstream.

MODULE 1

MINI-MODULE 1D

HOW THIS MODEL GUIDES TESTING & TREATMENT

Translating the HPATG Framework into Clinical Strategy:
Hypothesis, Testing, and Sequencing

THE CLINICAL PROCESS



SYMPTOM CLUSTER

EXAMPLE 1: FATIGUE

CLUSTER

Fatigue + Anxiety + PMS

HYPOTHESIS

High Cortisol stealing Progesterone

TEST

DUTCH Complete (Adrenals + Sex Hormones)



HPATG HYPOTHESIS

EXAMPLE 2: HAIR LOSS

CLUSTER

Hair Loss + Cold + Brain Fog

HYPOTHESIS

Thyroid Conversion Issue (Low T3)

TEST

Full Thyroid Panel (TSH, fT4, fT3, rT3, Abs)



TARGETED TEST

EXAMPLE 3: ACNE

CLUSTER

Acne + Missed Periods + Sugar Cravings

HYPOTHESIS

Insulin driving Androgens (PCOS)

TEST

Fasting Insulin + Glucose + Testosterone

"Don't guess. Hypothesize, then verify."

SEQUENCING INTERVENTIONS



WHY?

If you give thyroid meds to a patient with adrenal fatigue, you may cause anxiety. If you give estrogen to a patient with insulin resistance, you may worsen inflammation.

"Fix the foundation first. The roof comes last."

ANTICIPATING COMPENSATIONS

SCENARIO 1: CONSERVATION MODE

Adrenals Exhausted



Body Enters Conservation



Thyroid Downregulates

Energy rationing — a survival response

When the **Adrenals** are exhausted, the body slows the **Thyroid** to conserve energy. Treating the thyroid alone is like whipping a tired horse.

SCENARIO 2: THE STRESS SEESAW

Progesterone



*Luteal phase defect
Buffer is gone*

Cortisol

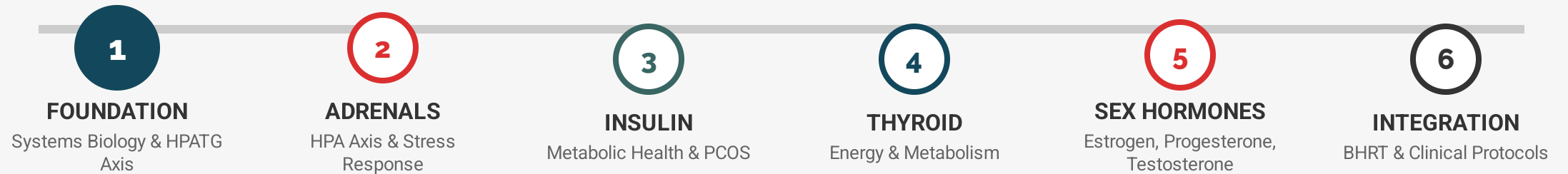


*Rises unchecked
Anxiety & insomnia*

When **Progesterone** drops, the buffer against stress disappears. **Cortisol** rises unchecked — the seesaw tips and stays tipped.

"The body never makes a mistake. It makes a compensation."

YOUR LEARNING JOURNEY



"We build the house from the ground up."

KEY TAKEAWAYS



SYSTEMS, NOT SYMPTOMS

The HPATG axis is your map. Don't get lost in the weeds of isolated symptoms.



SEQUENCE MATTERS

Treat the foundation (Brain → Adrenals → Thyroid) before the roof (Sex Hormones). Order is everything.



ANTICIPATE COMPENSATION

Always ask: "If this system is down, what is overworking to compensate?"

CLINICAL MASTERY

"The best protocol is the one that respects the body's hierarchy."



UP NEXT — MODULE 2

The Practitioner's Toolkit: How the HPATG Framework Guides Your Testing Strategy

MODULE 1

MODULE 1 COMPLETE

THE FOUNDATION IS SET

The HPATG axis is now your lens. Welcome to systems-based hormone medicine.